



SOGx2 EVB StarterKit-QSG

Part Number	902-SOG52-Vx0RH / 902-SOG72-Vx0RH
Platform Form	MediaTek MT8371 / MT8391
Product Name	SOG52 & SOG72 EVB StarterKit
Description	SOG52/ SOG72 Evaluation Kit based on MediaTek Genio 520/720
Revision	
Issue Date	

Approved by	Reviewed by	Issued by
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Revision History

Version	Date	Description
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1	2026/5/15	Release

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Packing List

No.	Part Number	Item Name	Description	Quantity
1	902-SOG52-Vx0RH / 902-SOG72-Vx0RH	SOG52/72 EVK Main Board	ASSY BOARD SOG52/SOG72 EVK Main Board	1
2	314-SSBE8-US0A8	12V DC JACK Power Adapter	AC ADAPTER 12V/5A + AC POWER CORD C13 US (T152)	1
3	430-SSBE8-131EF	IoT-CAM-DTB-D8V2	CAMERA MODULE D8 V2 (IMX258)	1
4	430-SSBE8-051EF	IoT-CAM-DTB-D9V2	CAMERA MODULE D9 V2 (OV5640)	1
5	007-SSBE8-C00EG	CAM-DTB-FPC-V2	CAMERA MODULE FPC L=250mm FOR SSBE8	2
6	306-SOG52-000XX	IPEX wire	COAXIAL CABLE SMA TO IPEX4 OD1.32*L200mm	2
7	087-SSBE6-W001G/ 087-UT180-A00EP	Antenna Cable set	SMA ANTENNA cable set	2
8	309-SOG52-000ER	USB CABLE TYPE-C TO TYPE-A	USB A to C CABLE L=1M (Debug cable not for charging)	1
9	007-SSBE8-W00A6	AW-XB468NF	WiFi 6 module-PCIe AW-XB468NF (7921)	1

SOGx2 SoM EVK System Setup

The SOG52/72 SoM EVK includes the following peripherals:

- SOG52/72 EVB Main Board * 1 Set
- D8 Camera Board * 1 Set
- D9 Camera Board * 1 Set
- Camera FPC * 2 Sets
- Wi-Fi Module With 2 Antennas * 1 Set
- 12V DC JACK Power Adapter * 1 Set
- USB Cable Type-C To Type-A * 1 Set (Used for Data transfer)



SOGx2 SoM EVK Overview

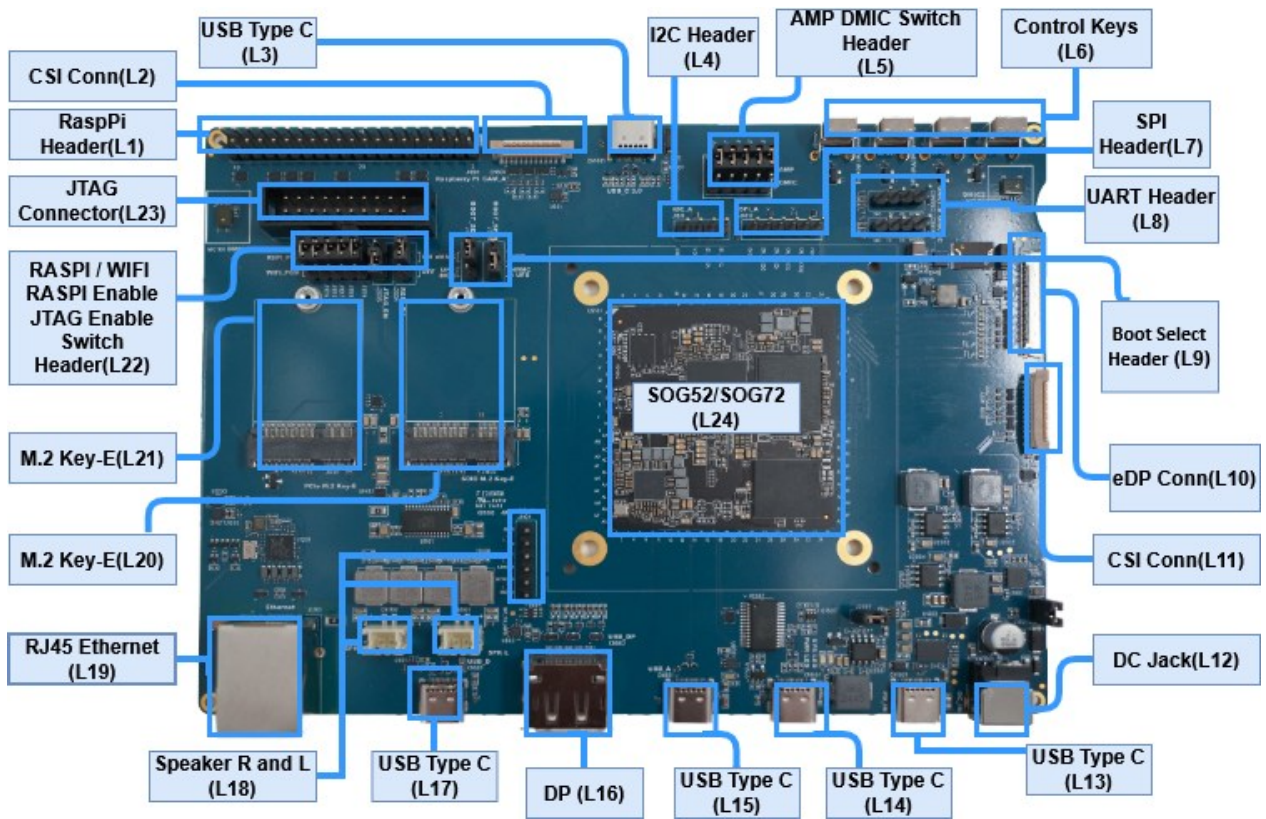


Figure 1 Top view

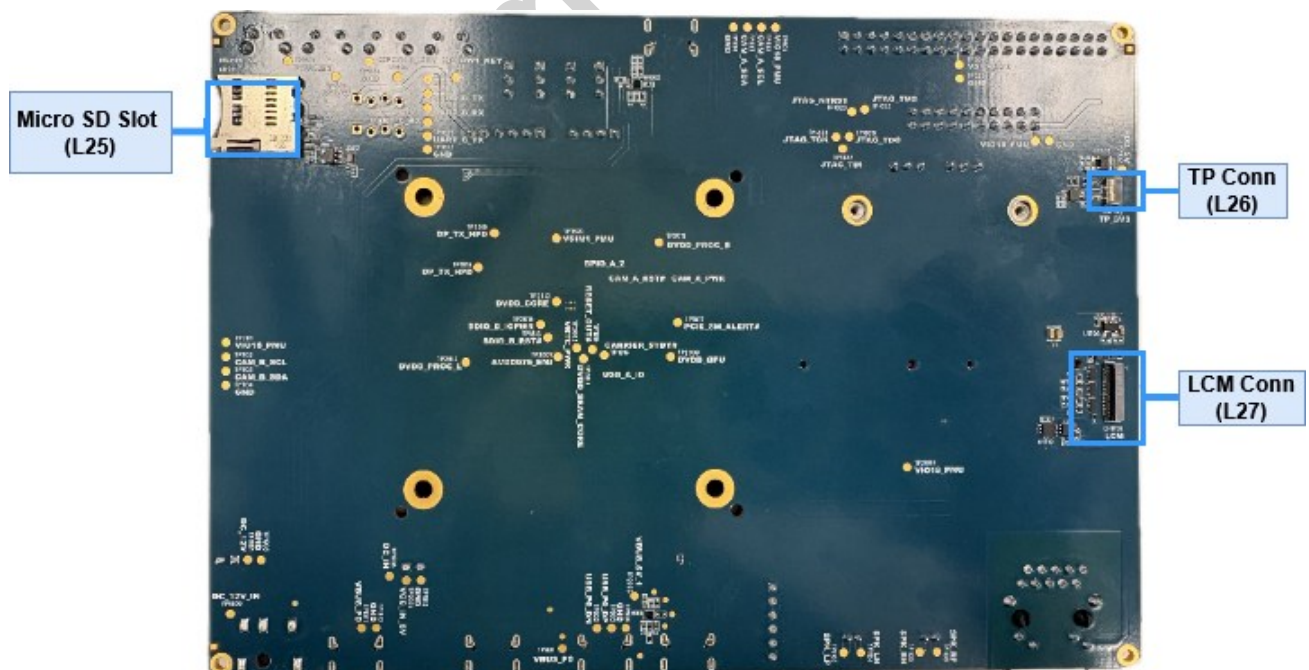


Figure 2 Bottom

Camera Board Installation

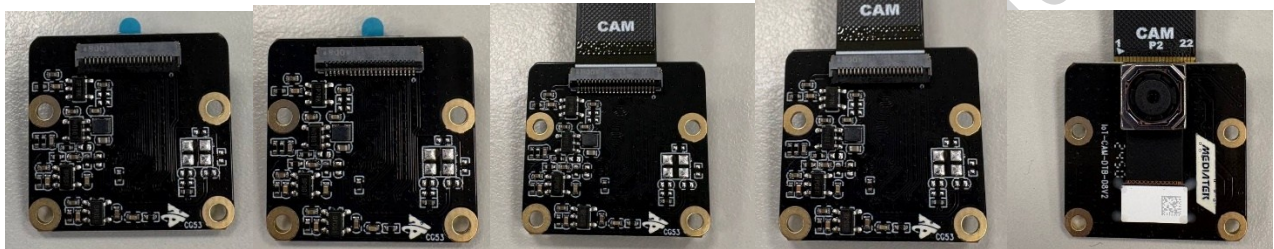
Camera Board Installation

Connect the D8/D9 camera board to the CAM0 or CAM1 connector on the SOGx2 SoM EVK using the camera FPC cable.

Installation procedure:

- Step 1

Loosen the camera board connector, insert the FPC into the camera board connector, and then lock it.



- Step 2

Loosen the connector on the SOGx2 SoM EVK, insert the camera FPC into the connector, and then lock it.

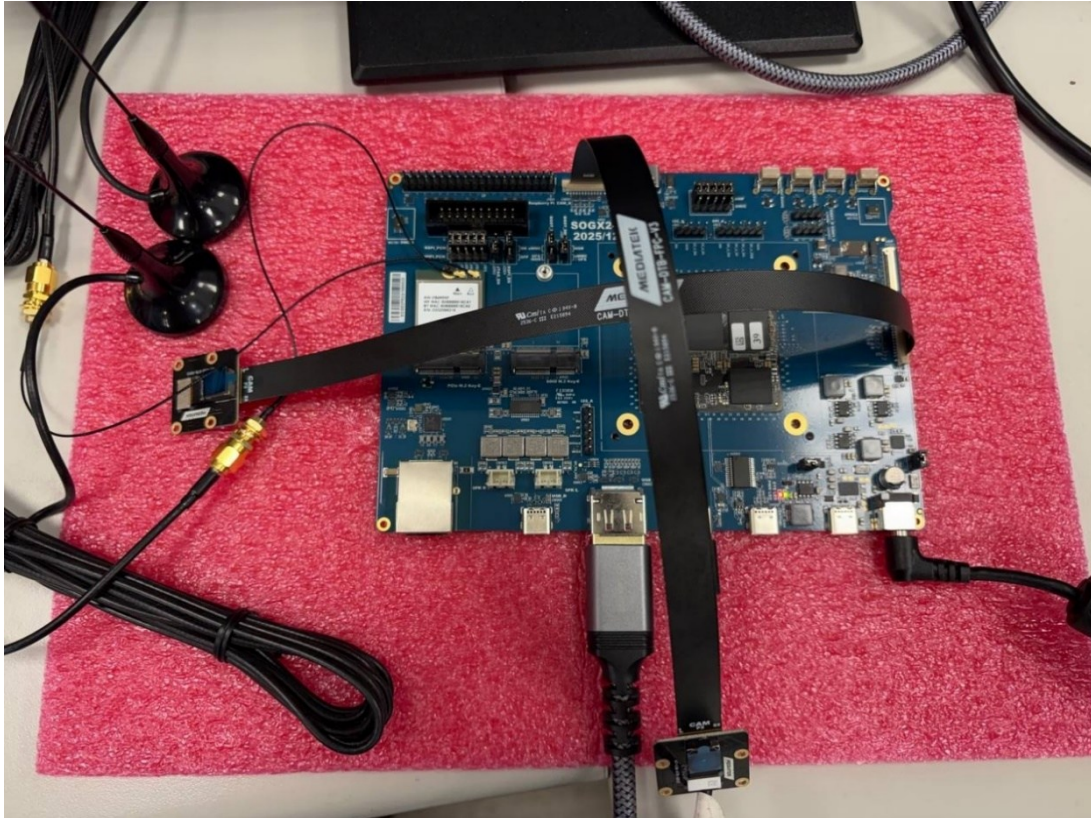




WiFi Module Installation

Insert the Wi-Fi module into the PCIe M.2 Key-E slot (L21), secure it with the locking screw, and install the Wi-Fi antennas.

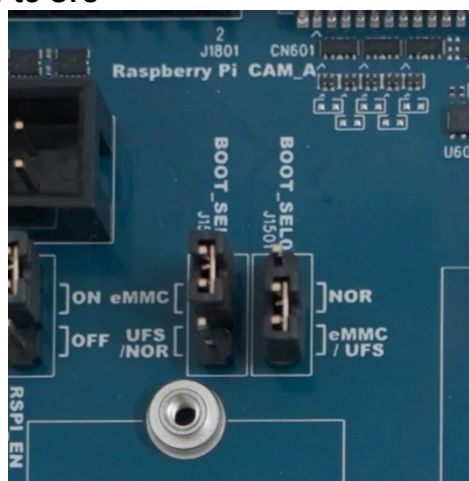




Boot Mode Select Jumper Configuration

Before proceeding with any boot or flashing operation, always confirm the target ROM (storage) device.

- Default: The SOG52/SOG72 EVK uses eMMC as the primary boot and storage device.
- If using a storage device other than eMMC, set the boot selection jumper (L9) accordingly.
 - NOR Flash: Set jumper to NOR
 - UFS: Set jumper to UFS



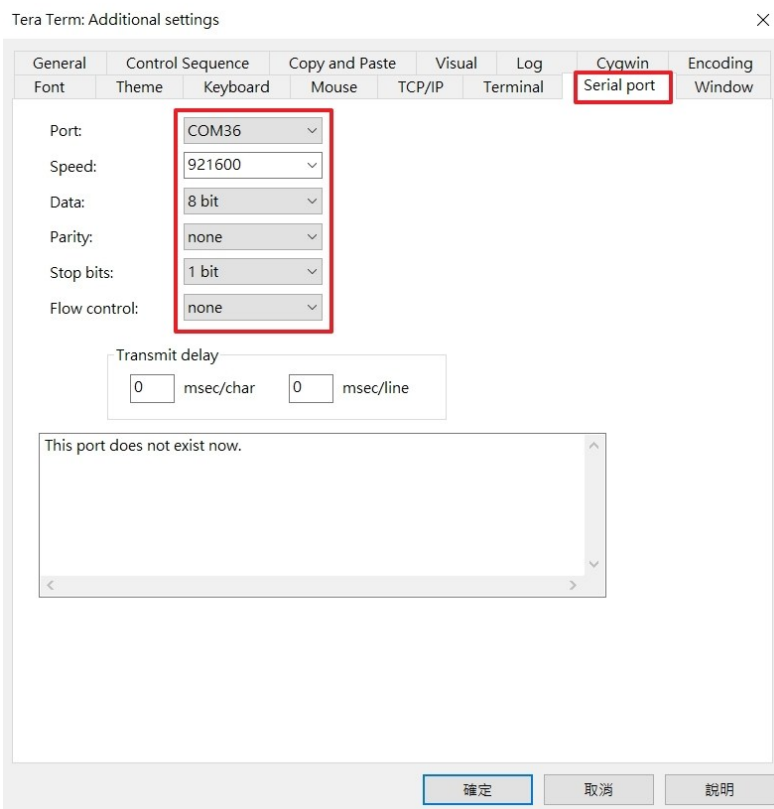
Function	Description		
Location	L9 (J1502 & J1501)		
Boot Mode	J1502 Setting (BOOT_SEL1)	J1501 Setting (BOOT_SEL0)	Description
NOR Flash	Lower (Pin 2-3)	Upper (Pin 1-2)	System boots from NOR Flash.
eMMC (Default)	Upper (Pin 1-2)	Lower (Pin 2-3)	System boots from eMMC.
UFS	Lower (Pin 2-3)	Lower (Pin 2-3)	System boots from UFS.

For all other jumper configurations and functions(AMP DMIC Switch & RASPI / WIFI & RASPI Enable & JTAG Enable Switch etc.), refer to the SOGx2-SOM-Carrier Board Hardware User Guide.

Installation procedure for viewing EVK logs

- Step 1: Connect UART0(L14) to the computer via a USB Type-C cable.
- Step 2: Plug in the DC JACK Power adapter.
- Step 3: Open serial tool such Tera Term, Putty....etc.
- Step 4: To configure the UART port settings, specify the following parameters:
Serial Port Number: Select the appropriate serial port number
Baud Rate: Set the baud rate to 921600
Data Bits/Parity/Stop Bits: Configure as 8 data bits, None parity, and 1 stop bit

You will see the screen shown on display and some trace logs coming out from UART0.



```
COM36 - Tera Term VT
File Edit Setup Control Window Help
6 [wifi.active.interface]=[20421 [wlan.driver.status]=[unloaded]20422 [wifi.active.interface]=[20423 [wlan.driver.status]=[
ok]20430 [init.svc.logd-auditctl]=[stopped]20469 Done
[ 20.481324][ T704] [DISP][E]layering_rule_start error:-14
[ 20.497975][ T1] init 22: Service 'exec 26 (/system/bin/lmkd --boot_completed)' (pid 1749) exited with status 0 oneshot
service took 0.079000 seconds in background
[ 20.498832][ T704] [DISP]HRT hrt_num:0xffffffff/disp_idx:0/disp_list:0/mod:0/dal:0/addon_scn:(0, 0, 0)/bd_tb:0/i:0
[ 20.500131][ T1] init 22: Sending signal 9 to service 'exec 26 (/system/bin/lmkd --boot_completed)' (pid 1749) process
group...
[ 20.501215][ T704] [DISP][E][HRT] gles invalid, disp:0, head:0, tail:2
[ 20.503317][ T1] libprocessgroup 25: Removed cgroup /sys/fs/cgroup/systemd/0/pid_1749
[ 20.503529][ T704] [DISP][E]check_disp_info fail
[ 20.503533][ T704] [DISP][E]layering_rule_start error:-14
[ 20.545561][ T1] init 22: processing action (sys.boot_completed=1) from (/system_ext/etc/init/batterywarning.rc:8)
[ 20.547495][ T1] init 29: starting service 'batterywarning'...
[ 20.548083][ T704] [DISP]HRT hrt_num:0xffffffff/disp_idx:0/disp_list:0/mod:0/dal:0/addon_scn:(0, 0, 0)/bd_tb:0/i:0
[ 20.549603][ T704] [DISP][E][HRT] gles invalid, disp:0, head:0, tail:2
[ 20.550441][ T704] [DISP][E]check_disp_info fail
[ 20.551040][ T704] [DISP][E]layering_rule_start error:-14
[ 20.556637][ T1] init 22: ... started service 'batterywarning' has pid 1776
[ 20.557670][ T1] init 22: processing action (sys.boot_completed=1) from (/vendor/etc/init/atcid.rc:17)
[ 20.559983][ T1] init 22: processing action (ro.build.type=userdebug && sys.boot_completed=1) from (/vendor/etc/init/mt
klog.rc:3)
[ 20.562171][ T1] init 22: Command 'write /proc/dynamic_debug/control file *mediatek* +p ; file *gpu* =' action=ro.buil
d.type=userdebug && sys.boot_completed=1 (/vendor/etc/init/mtklog.rc:4) took 0ms and failed: Unable to write to file '/proc/d
ynamic_debug/control': open() failed: Permission denied
[ 20.565512][ T704] [DISP]HRT hrt_num:0xffffffff/disp_idx:0/disp_list:0/mod:0/dal:0/addon_scn:(0, 0, 0)/bd_tb:0/i:0
[ 20.565523][ T704] [DISP][E][HRT] gles invalid, disp:0, head:0, tail:2
[ 20.565529][ T704] [DISP][E]check_disp_info fail
[ 20.565532][ T704] [DISP][E]layering_rule_start error:-14
[ 20.569367][ T1] init 22: processing action (sys.boot_completed=1) from (/vendor/etc/init/mtklog.rc:6)
[ 20.571026][ T305] log_store: sram_dram_buff flag 0x223, reboot count 0, 0.
[ 20.572216][ T305] log_store:read pmic register value 0x100.
[ 20.572949][ T305] log_store: write pmic value 0x1400, ret 0x0.
[ 20.573707][ T305] log_store: printk_buff addr:0x223b10000,sz:0x80000,buff-flag:0x623.
[ 20.575054][ T305] console name: ttyS, status 0x16.
```

